NLP Group Assignment 2

Combined technical report - Questions 1 to 4

DSL 504 - Natural Language Processing
Deadline: 30 October 2026, 11:59 PM IST
Coordinator and final submitter: Arnav Mishra

Member	ID	Primary question
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Section	Combined PDF pages	Content
Question 1	2-6	Models, curves, length analysis and six attention/error cases
Question 2	7-12	Six-page section; completed training/evaluation; seven comparative examples
Question 3	13-23	Revised experiment, generation attention, qualitative analysis and limitations
Question 4	24-31	Models, decoding, generated stories, quantitative results and limitations

The Q2 section is exactly six pages, including its figures and tables. There is no stated six-page cap for the whole group report. Question-specific section reports remain alongside the source; this PDF presents the complete four-question technical narrative. The question-level documents contain their experiment-specific references and limitations.

The contribution split and question responsibilities are listed in CONTRIBUTIONS.md.

Question 1: Attention-based English-Hindi translation

Owner: Rudra Mehul Dudhat. Two LSTM encoder-decoder systems compare fixed-state decoding with manually implemented additive attention. Results are documented by the executed notebook, saved histories, translations and attention figures.

1. Data preparation and architecture

The IIT Bombay corpus is normalized and tokenized, then empty/malformed pairs, excessive length/ratio and non-Devanagari targets are filtered. From 200,000 sampled raw training pairs, 50,000 cleaned pairs are used. Training pairs have at most 25 tokens per side; retained development/test pairs have at most 30. The filtered evaluation contains 430 development and 1,723 test pairs. Each language vocabulary contains 10,000 entries with PAD/SOS/EOS/UNK, fitted on training only. Maximum training length is 27 including boundary tokens; Hindi training OOV rate is about 9.8%.

Setting	Baseline	Attention
Architecture	128d embedding; 256d LSTM encoder/decoder; final state initializes decoder	Same sizes; additive scores over all encoder states; masked softmax context feeds decoder
Approx. parameters	5.9 million	6.3 million
Training (both)	Adam lr 0.002; batch 128; dropout 0.3; seed 0	8 epochs; teacher forcing; clip 1.0; validation-loss-selected checkpoint
Decoding (both)	Greedy; maximum 40 steps	PAD/SOS and UNK suppressed as disclosed

Attention uses v^T tanh(W_s s + W_h h_i), excludes padded source positions and forms a weighted encoder-state context. This exposes token-level weights without relying on a prebuilt translation model. The models share major training conditions, but attention adds parameters. Both were trained concurrently on CPU in the supplied run; epoch times are not isolated speed benchmarks.

Measured result	Baseline	Attention
Best validation loss	4.760	4.654
Teacher-forced validation next-token accuracy	0.289	0.302
Autoregressive test corpus BLEU (0-100)	0.60	0.92
Test word-overlap F1	0.187	0.205

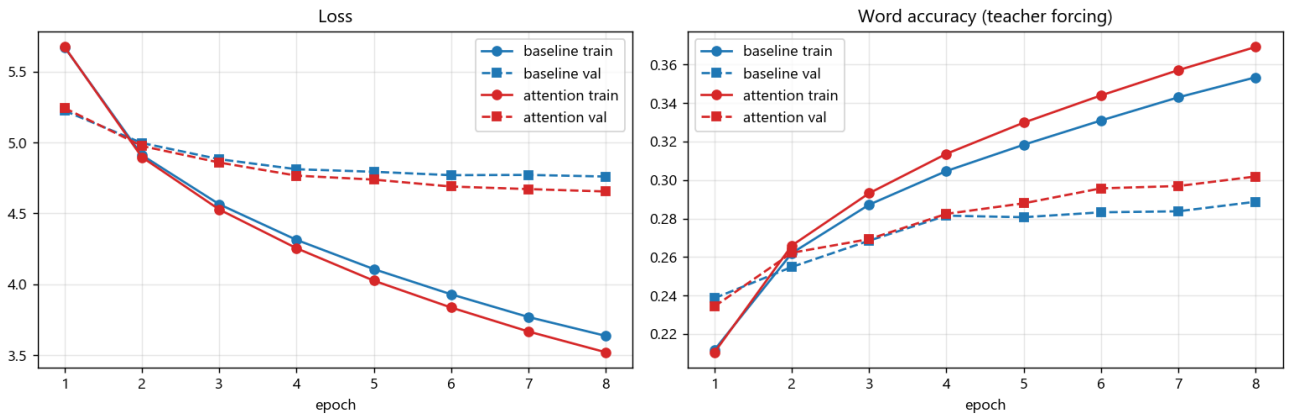


Figure Q1.1. Supplied training/validation losses and token accuracies. Teacher-forced accuracy supplies gold preceding tokens; it is not generated-translation accuracy. Both models remain weak under autoregressive decoding.

2. Length, errors and what attention explains

Source tokens	Test count	Baseline BLEU	Attention BLEU
1-5	10	1.97	2.50
6-10	256	1.25	1.14
11-15	455	0.56	1.09
16+	1,002	0.49	0.82

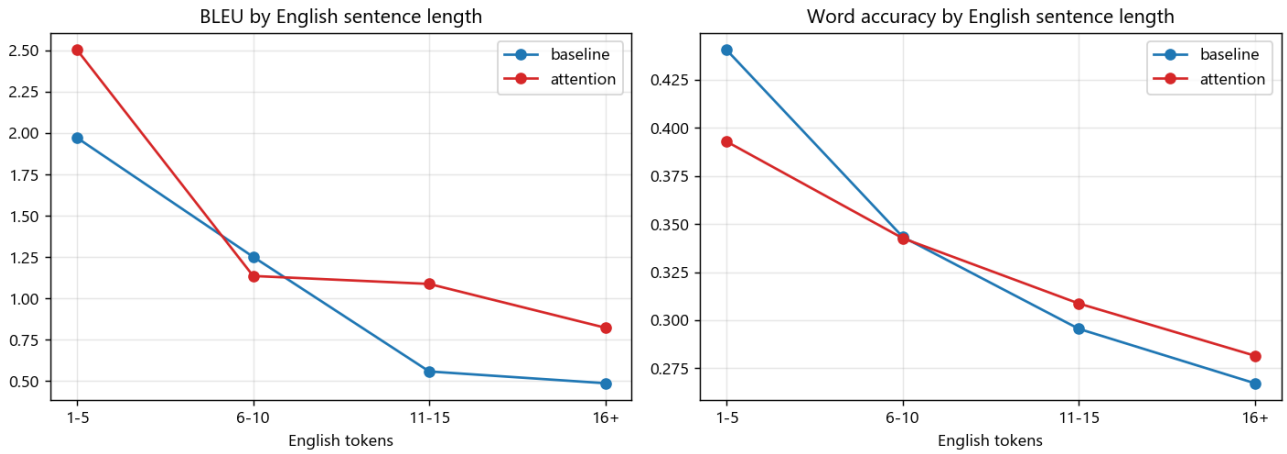


Figure Q1.2. Supplied length analysis. The accuracy subplot is teacher-forced next-token accuracy; BLEU uses generated translations. The shortest bucket has only ten sentences, so its apparent gain is uncertain. Attention improves BLEU in the 11-15 and 16+ buckets, while the 6-10 bucket is slightly worse. Word-overlap gains do not increase monotonically with length.

Attention improves word overlap on 926 test sentences, worsens it on 718 and ties on 79. Mean paired gain is 0.0178 with reported sentence-level standard error 0.0024; this is one seed per architecture and does not quantify training-seed uncertainty. Heuristic missing-word flags decrease from 43.0% to 38.2%, while repetition increases from 37.1% to 46.3% and extra-word flags from 29.5% to 34.3%. These rule-based categories are diagnostic indicators, not validated linguistic annotations.

The historical 30k-pair run used six epochs, vocabulary 8k and batch 64; the current run uses 50k/eight epochs/vocabulary 10k/batch 128. Its higher scores cannot be causally attributed to more data alone. The historical curves/translations are retained as supporting evidence rather than a controlled data-size ablation.

The supplied attention analysis reports a mean peak weight of 0.32, 86% of steps with no source position above 0.5 and sharpness/word-overlap correlation 0.20. Looking at a relevant source word can still yield an incorrect or incomplete translation. The first/last-third recall statistic concerns Hindi reference position; because word order differs across languages, it does not establish loss of the English source prefix. Likewise, a >=16-token input alone does not diagnose a long-distance dependency error. The six cases below describe observed omissions, substitutions and repetition without making that automatic causal claim.

Limitations: restricted noisy corpus, word-level vocabularies, small development set, only one model seed, greedy decoding and UNK suppression; the latter hides unknown-token outputs rather than recovering missing information. Neither BLEU below 1 nor visually sharp attention establishes useful translation. Longer training, subwords and other decoding are future options, not verified improvements in this run.

Sources: Q1 REPORT.md, executed notebook, saved histories, test_translations.csv and figures.

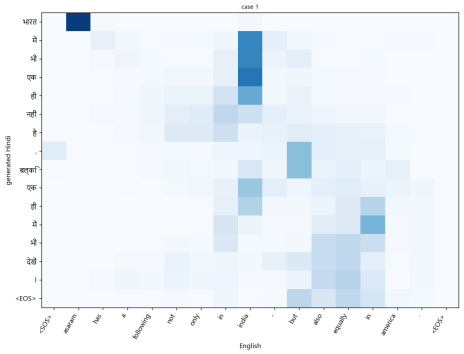
3. Attention error cases 1-2

Case 1. English: asaram has a following not only in india , but also equally in america .

Reference: आसाराम का भारत में ही नहीं , बल्कि अमेरिका में ही उतना भी प्रभुत्व है ।

Baseline: भारत में इस प्रकार की एक ऐसी हालत जो भी नहीं है ।

Attention: भारत में भी एक ही नहीं है , बल्कि एक ही में भी देखें ।



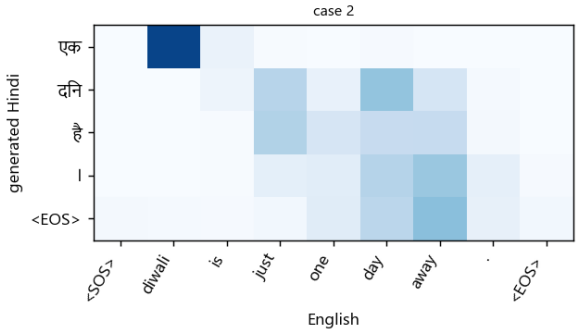
The unknown name Asaram is replaced; America is omitted. The contrastive connector is partly retained. Sharp focus on an OOV source name does not supply its target translation.

Case 2. English: diwali is just one day away .

Reference: दीपावली को एक दिन शेष बचा है ।

Baseline: एक व्यक्ति जो कुछ भी हो ।

Attention: एक दिन है ।



The generated text keeps one day but omits Diwali and the remaining-time relation. The model attends to an unknown event word while producing a generic phrase.

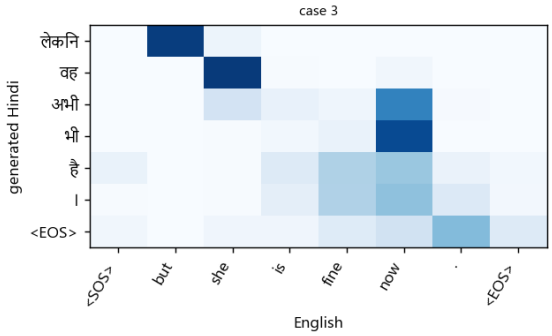
3. Attention error cases 3-4

Case 3. English: but she is fine now .

Reference: लेकिन अब वह अच्छी है ।

Baseline: लेकिन यह सच है ।

Attention: लेकिन वह अभी भी है ।



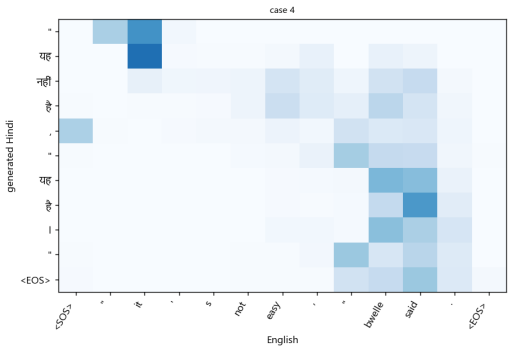
But/she/now are partly aligned, but the key property fine is absent. A plausible alignment is therefore insufficient for a complete translation.

Case 4. English: " it ' s not easy , " bwelle said .

Reference: वेल ने कहा , " यह आसान नहीं है । "

Baseline: " मैं तुम्हें नहीं है , " मैं तुम्हें नहीं है ।

Attention: " यह नहीं है , " यह है । "



The reporting name is unknown and output repeats fragments. Repetition and an incomplete quotation are directly observed; the heatmap is a clue rather than a causal proof.

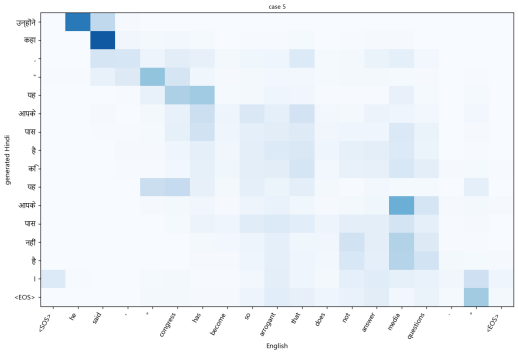
3. Attention error cases 5-6

Case 5. English: he said , " congress has become so arrogant that does not answer media questions . "

Reference: उन्होंने कहा , " कांग्रेस इतनी अहंकारी हो गई है कि वह मीडिया के सवालों का जवाब नहीं देती है । "

Baseline: उसने कहा , " ऐ मूसा ! " तुम कह रहा है , " तुम सच्चे हो , " तुम लोग गुमराही की

Attention: उन्होंने कहा , " यह आपके पास है कि यह आपके पास नहीं है ।



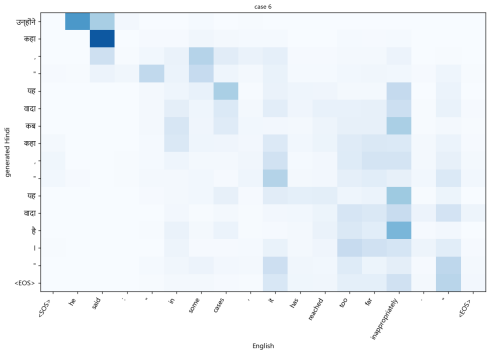
The reporting phrase survives, while the Congress/arrogance/media-answer content is lost or replaced. The long source alone is not proof of a specific dependency failure.

Case 6. English: he said : " in some cases , it has reached too far inappropriately . "

Reference: उन्होंने कहा किः " कुछ मामलों में , यह अनुपयुक्त रूप से बहुत आगे चली गई है । "

Baseline: उसने कहा , " हम तो एक ही एक अनुस्मृति है , " मेरे पास लाओ था ।

Attention: उन्होंने कहा , " यह वादा कब कहा , " यह वादा है । "



The reporting phrase survives but the quote becomes repetitive and semantically unrelated. OOV content and diffuse attention are plausible contributors, not isolated causes.

Question 2: Five-class Amazon review rating prediction

BiGRU with pretrained GloVe and additive attention versus fully fine-tuned RoBERTa with masked mean pooling. Experiments use seeds 42 and 123; first-token pooling is the selected ablation. This section reports inspected saved measurements, without additional model selection after final test evaluation.

1. Data, preprocessing and controlled training

The official SetFit English split contains 200,000 training, 5,000 validation and 5,000 test reviews, balanced across stars 1-5. SetFit supplies bodies but omits titles. Titles were recovered from the original-corpus mirror using exact agreement of ID, body, language and stars, then joined as title + newline + body for both models. Null titles (33 train, one validation) remain empty. Vocabulary and preprocessing fit training only; validation chooses checkpoints. Five exact train/validation text overlaps are disclosed while retaining the required official splits. Reserved test access followed a frozen protocol.

The classic pipeline HTML-unescapes, lowercases and removes tags/URLs, then applies regex tokenization. Minimum frequency 2 yields 32,611 vocabulary entries, capped at 60,000, with PAD/UNK and a 200-token limit. GloVe 6B 100-dimensional vectors cover 87.77% of vocabulary; uncovered vectors use fixed seed 2026. RoBERTa uses original text, byte-level BPE and a 256-token limit with dynamic padding to multiples of eight. Dataset/model revisions and hashes are pinned in config.json and manifests.

Setting	BiGRU + attention	RoBERTa mean / first
Architecture	2 BiGRU layers; 128 hidden/direction; additive attention width 128	roberta-base; all encoder weights trainable; masked mean / first token
Head / parameters	Dropout 0.3; linear -> 5; 3,768,497	Dropout 0.2; linear -> 5; 124,058,885
Optimizer	Adam, learning rate 0.001	AdamW, learning rate 0.00002; decay 0.01; 8% warmup then linear decay
Batch / precision	128; float32	Global microbatch 16; accumulation 2; effective 32; recorded bfloat16
Selection / cap	Validation Macro F1; patience 2; at most 8 epochs	Validation Macro F1; patience 1; at most 3 epochs

Both use cross-entropy, gradient clipping 1.0 and minimum improvement 0.0001. GloVe is frozen for epoch one then unfrozen. Packed GRU sequences and masked attention exclude padding. RoBERTa mean pooling includes all non-padding states, including special tokens. AdamW excludes biases/LayerNorm from decay. All recorded runs used Tesla T4 hardware; optimizer/scheduler state was retained during active recovery.

Run	Seed	Epochs / best	Training h	Execution
baseline	42	7 / 5	0.22	1 GPU(s)
baseline	123	6 / 4	0.19	1 GPU(s)
roberta_mean	42	3 / 3	6.44	1 GPU(s)
roberta_mean	123	3 / 2	3.55	2 GPU(s)
roberta_first	42	3 / 3	3.56	2 GPU(s)
roberta_first	123	3 / 2	3.43	2 GPU(s)

2. Learning curves and selection

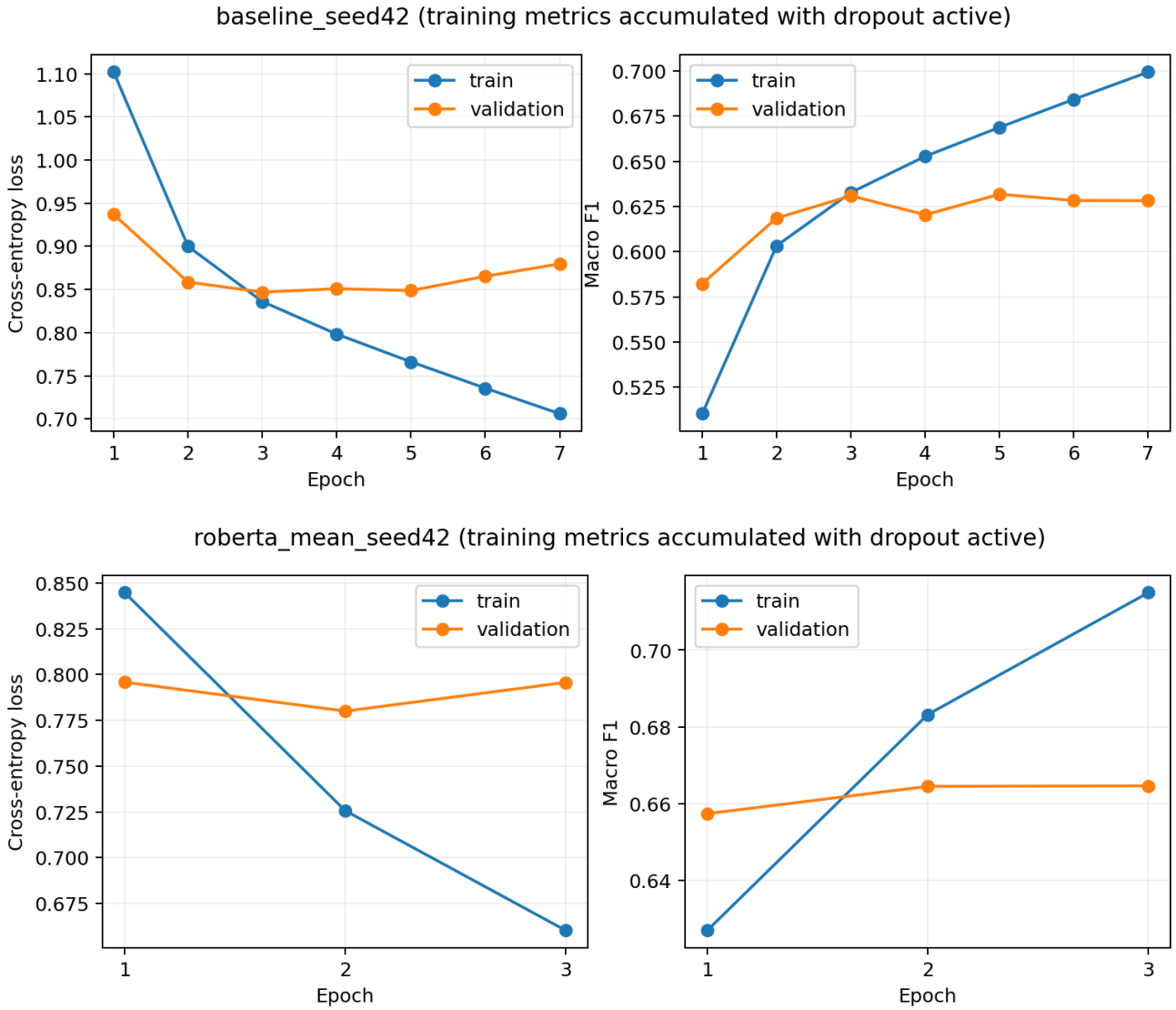


Figure 1. Training/validation loss and Macro F1 for seed 42 of the two main models. Equivalent figures for seed 123 and both ablation seeds are supplied in results/figures/. Training metrics accumulate with dropout active; validation uses evaluation mode, so the two curves are not identical evaluation protocols.

Baseline best epochs were 5/4 for seeds 42/123, with early stopping at epochs 7/6. Mean-pooling RoBERTa selected epochs 3/2; both runs completed the three-epoch cap. Checkpoints use validation Macro F1, not minimum loss or test accuracy. Full histories retain the actual losses and scores rather than only the selected epochs.

3. Frozen final test evaluation

Each main model/seed produced one final 5,000-review test prediction file. Inspection checked sample IDs, probability ranges/sums, argmax labels and recomputed metrics, including aggregate means and sample SD (ddof=1). Predictions use argmax with labels 0-4 mapped to stars 1-5. First-token models are evaluated on validation for the ablation, not additionally on test.

Model	Accuracy (%)	Macro F1	MAE
BiGRU	63.79 +/- 0.78	0.6351 +/- 0.0069	0.4277 +/- 0.0075
RoBERTa mean	67.13 +/- 0.04	0.6706 +/- 0.0010	0.3672 +/- 0.0000

RoBERTa improves accuracy by 3.34 percentage points and Macro F1 by 0.0355, while reducing MAE by 0.0605 stars. These are descriptive two-seed comparisons, not confidence intervals or significance tests. Smaller observed seed variability does not establish robustness.

True rating	BiGRU F1, mean +/- SD	RoBERTa F1, mean +/- SD
1	0.7197 +/- 0.0032	0.7454 +/- 0.0085
2	0.5483 +/- 0.0072	0.5911 +/- 0.0080
3	0.5360 +/- 0.0107	0.5792 +/- 0.0059
4	0.5908 +/- 0.0129	0.6369 +/- 0.0002
5	0.7805 +/- 0.0007	0.8004 +/- 0.0055

RoBERTa improves F1 for every rating; the largest gain is for four stars (about 0.0460). Both models identify extreme ratings more accurately than two-, three- and four-star reviews. Middle-rating wording is often mixed and neighboring rating labels can be ambiguous.

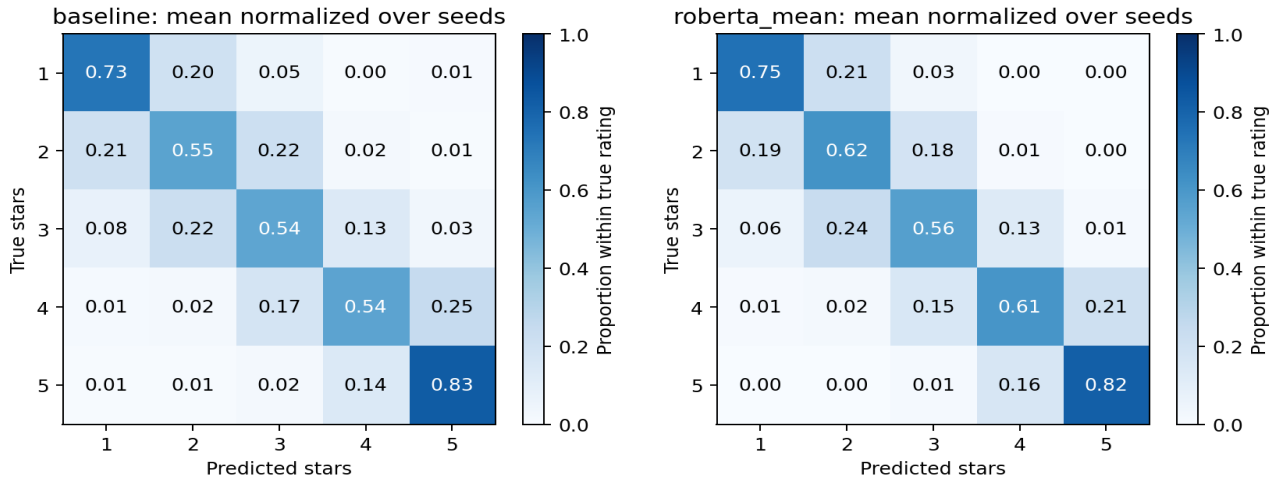
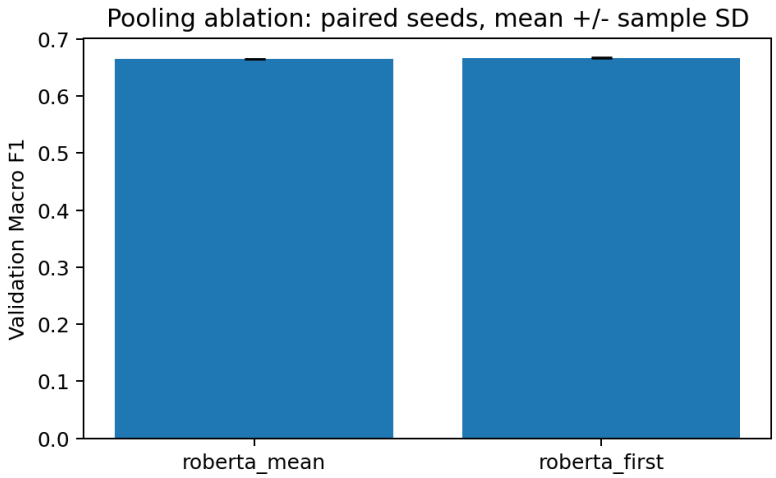


Figure 2. Row-normalized test confusion matrices averaged over seeds: BiGRU (left), RoBERTa (right). Full seed-specific matrices are supplied.

Adjacent-rating error rates fall from 30.84% to 29.49% of all reviews; distant errors (distance >=2) fall from 5.37% to 3.38%. Adjacent errors comprise 85.16% versus 89.72% of mistakes: the larger share for RoBERTa reflects fewer distant errors, not a higher overall adjacent-error rate. Extreme 1-versus-5 error rates fall from 0.29% to 0.09%.

4. Pooling ablation, reproducibility and limitations

Validation result	Mean pooling	First-token pooling
Macro F1 mean +/- SD	0.664875 +/- 0.000313	0.666848 +/- 0.000459
MAE mean +/- SD	0.3738 +/- 0.0006	0.3725 +/- 0.0016



First-token pooling is slightly better on validation in both seeds: mean-minus-first Macro F1 differences are -0.001869 (seed 42) and -0.002076 (seed 123). Its average MAE is lower by 0.0013. These small differences do not establish a universal pooling advantage. The required mean-pooling architecture remains the main final-test model; the ablation does not trigger test-driven architecture replacement.

Execution topology limits pooling-only attribution. Mean seed 42 used one GPU, while mean seed 123 and both first-token runs used two-GPU DistributedDataParallel. Global microbatch 16, accumulation 2 and effective batch 32 were preserved; gradient/communication probes passed, but rank-specific dropout streams and numerical reduction order can alter trajectories. The seed-123 pair shares topology; the seed-42 pair does not. Both baselines used one GPU. Actual topology, precision and source provenance are retained per run.

Training and evaluation were performed in the recorded Kaggle environment; CPU finalization only inspected and packaged saved files. It did not retrain or access the test corpus. Six training archives retain the selected checkpoints; the compact submission component excludes full data/caches/checkpoints. The final test protocol hash is recorded in FROZEN_PROTOCOL.json, and generated analysis can reuse its saved predictions.

Limitations include only two seeds, a single review domain, rare/OOV tokens, bounded sequence lengths, five known train/validation text overlaps, and ambiguity between neighboring star labels. RoBERTa also benefits from extensive external pretraining, so this comparison does not isolate architecture alone. No calibration, sarcasm understanding, deployment safety or out-of-domain robustness was established. Improvements should be selected on new training/validation experiments, not the final test results.

Conclusion: RoBERTa mean pooling improves the measured rating-classification and ordinal-error metrics over this GloVe-BiGRU baseline. First-token pooling has a small validation advantage in the present ablation. Neither finding establishes a general superiority across datasets, execution environments or untested hyperparameters.

5. Baseline-wrong / RoBERTa-right examples

Fixed seed 42, selected by ascending review ID: five of 500 available baseline-wrong/RoBERTa-right cases. Labels and predictions are unedited; interpretation below is a text-supported hypothesis, not a causal account of internal reasoning. Full texts are also retained in qualitative_examples.csv.

en_0002917: true 3; BiGRU 4; RoBERTa 3

Title: I also think I need to try a different coffee because I haven't been real pleased with the flavor of the coffee
Review: I've done a few so far but getting the hang of securing the foil to the capsule takes patience and practice. I've put holes in a few foils and had to redo them. I also think I need to try a different coffee because I haven't been real pleased with the flavor of the coffee. Also, the original foils on the capsules can be difficult to remove.

Mixed sentiment and practical difficulties. The review combines partial progress with flavor dissatisfaction, damaged foils and difficult capsule preparation. The neutral three-star label fits this mixed evidence. RoBERTa may better combine these qualifications; the positive four-star baseline prediction underweights complaints.

en_0003912: true 3; BiGRU 2; RoBERTa 3

Title: I won’t wear it much
Review: It’s a bit expensive and doesn’t lay right, it sticks to everything.

Rating ambiguity and malformed text. The text is strongly critical, but its recorded label is three stars. RoBERTa matches that label while the baseline predicts two. This is label agreement, not proof that three stars is the uniquely correct interpretation. The malformed apostrophes are present in the saved text.

en_0005862: true 2; BiGRU 3; RoBERTa 2

Title: Loved this purse but
Review: Loved this purse but, I can't take the bad odor coming from inside the purse. I even aired it out for two days and there was no difference. In process of returning.

Contrast and overriding complaint. The positive opening is reversed by persistent odor, unsuccessful airing and a return. RoBERTa predicts two stars while the baseline predicts three; sensitivity to the contrast and return intent is a plausible explanation, not a demonstrated reasoning process.

en_0008881: true 2; BiGRU 3; RoBERTa 2

Title: ... Expected it to be made out of a material like a bath towel
Review: Too Warm - Expected it to be made out of a material like a bath towel. Very soft, but seems like an old lady would wear this to stay warm inside a building that the A/C is on in the summertime on a 95 degree day...

Mixed attributes and unmet expectations. Softness is positive, but excessive warmth and the mismatch with expected towel-like material dominate the complaint. RoBERTa predicts two stars; the baseline predicts three. This suggests different weighting of positive attributes and dissatisfaction.

en_0009772: true 2; BiGRU 3; RoBERTa 2

Title: Could not find the answers I needed
Review: The Puzzle dictionary is too heavy to handle. The words, Clues, I tried to look up were more difficult than the clue itself. Would not suggest it for making Crossword Puzzles easier.

Negative utility and recommendation. The dictionary is described as heavy, unhelpful and difficult to use, with an explicit negative recommendation. RoBERTa matches two stars while the baseline gives three. This example concerns failure of the intended use rather than sarcasm.

6. Reverse cases and token-level importance

Two of 362 available baseline-right/RoBERTa-wrong cases, again in ascending review-ID order for seed 42. The stronger aggregate model still fails on individual reviews.

en_0000916: true 3; BiGRU 3; RoBERTa 2

Title: Nice mat but slips all over the place
Review: Doesn't stay put. Annoying.

Positive title versus short negative body. The title calls the mat nice but the body says it slips and is annoying. The baseline matches three stars; RoBERTa predicts two. The latter may overemphasize the complaint. The short mixed review illustrates uncertainty in mapping wording to exact stars.

en_0005599: true 5; BiGRU 5; RoBERTa 4

Title: Very classy and very sturdy
Review: We bought this to use in our RV. We like it very much. Very classy and very sturdy (so far). It is not light weight (ie, like Correlle) but we were not necessarily looking for anything lightweight). However, is is very attractive and so far we have enjoyed using it in our RV.

Qualification within strong approval. The reviewer repeatedly likes the product and accepts its weight as compatible with their needs. The baseline matches five stars; RoBERTa gives four. Mention of weight or the qualification so far may contribute, but the explanation remains a hypothesis.

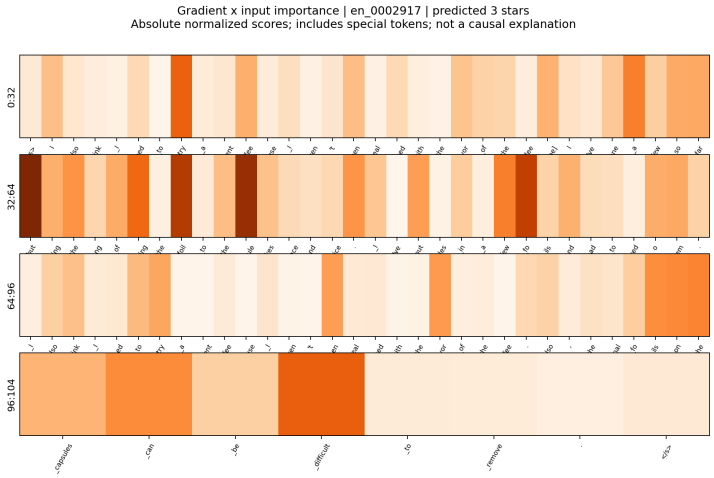


Figure 3. Absolute gradient x input importance for the predicted-class logit of the fixed RoBERTa model on en_0002917. Scores are normalized across all retained tokens, including special tokens. The displayed subset is not the complete review; full scores are in token_importance.csv.

In the saved scores, the contrast marker but (position 32) has importance 0.0398; capsule 0.0368, foil 0.0334, the fo subword of foils 0.0322, difficult 0.0264 and try 0.0261 also receive substantial weight. This sensitivity is consistent with practical difficulties and mixed sentiment, but does not establish how the model combined them or prove negation understanding. Tokens for the first occurrence of the negation have smaller scores than some context tokens. The example has 104 retained tokens, below the 256-token limit; truncation is therefore not the explanation for this particular difference. Other long reviews may lose text.

Reproduction/evidence: question_2/README.md and KAGGLE_RUN_GUIDE.md describe exact phases; config.json, manifests, environment records, training_records/, final prediction CSVs and metric JSONs retain run-level provenance. Qualitative interpretations are based on the saved review text and model predictions.

Sources: Assignment PDF pp. 5-7; SetFit/amazon_reviews_multi_en and buruzaemon original-corpus mirror (exact IDs/revisions in config.json); Stanford GloVe 6B; FacebookAI/roberta-base. Full source links and recorded settings are retained in the component README/configuration.

Abstractive Text Summarization

Question 3 - From-scratch Transformer Encoder-Decoder

Measured CPU experiment on CNN / Daily Mail 3.0.0. All reported model scores come from autoregressive generation with the validation-selected checkpoint; no pre-trained model weights are used.

1. Data and preprocessing

The official splits contain 287,113 training, 13,368 validation, and 11,490 test pairs. To fit CPU resources, this experiment uses the first 2,048 training pairs and first 1,024 validation pairs in repository order. The training subset is 0.71% of the full training split and is not a random representative sample. All 11,490 test articles are evaluated. No test example trains the tokenizer, model, or hyperparameters.

SentencePiece BPE is trained exclusively on lowercased articles and highlights from the training subset. One shared 4,096-token vocabulary is used for both sides, including PAD=0, UNK=1, BOS=2 and EOS=3. SentencePiece applies NFKC normalization. Source articles are truncated to 400 subwords. Summaries have at most 128 tokens including BOS and EOS (126 content subwords). Batches are padded to their longest sequence. Original reference highlight newlines are retained separately for scoring.

2. Architecture

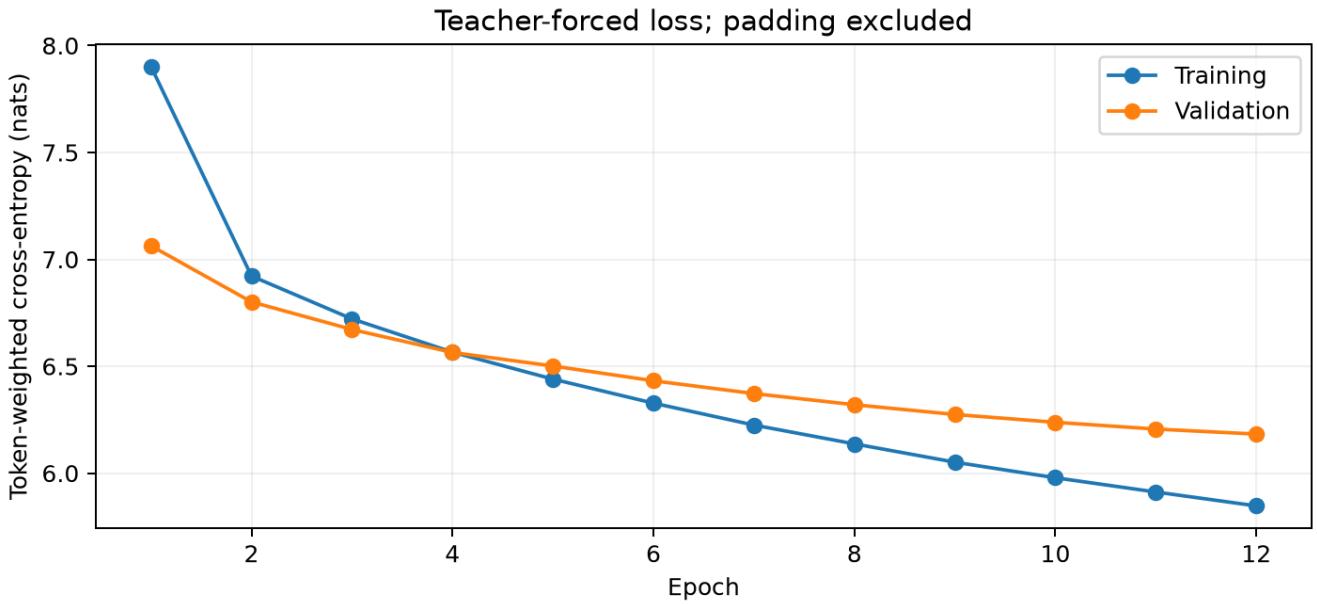
Setting	Value
Encoder / decoder layers	2 / 2
Model / feed-forward width	64 / 256
Attention heads	4 (16 dimensions each)
Dropout / normalization	0.1 / post-residual LayerNorm
Trainable parameters	761,856
Position / output layer	Sinusoidal encoding / linear vocabulary logits

Query, key and value projections are reshaped into heads. Attention is softmax(QK^T / sqrt(16) + mask)V, followed by concatenation and a linear projection. Feed-forward layers use Linear -> ReLU -> Dropout -> Linear. Each sublayer uses LayerNorm(x + Dropout(sublayer(x))). The source padding mask is used in the encoder and cross-attention. Decoder self-attention combines a triangular causal mask and target-key padding mask. All components use elementary tensor operations and basic neural layers.

3. Training and checkpoint selection

Teacher forcing uses gold summaries shifted right as decoder inputs and shifted left as targets. Cross-entropy ignores PAD; epoch losses are weighted by the number of non-padding target tokens. Adam uses betas=(0.9, 0.98), epsilon=1e-9, gradient clipping at 1.0, batch size 16 and seed 42. The learning rate is 0.1 * 64^(-0.5) * min(step^(-0.5), step * 200^(-1.5)). Early stopping has patience 2 after epoch-end validation; the best checkpoint is restored.

The run completed 12 epochs. Best epoch: 12; validation loss: 6.1827 nats/token. Training time: 15.7 minutes on CPU with 4 PyTorch threads. PyTorch 2.14.1+cpu. This width and depth are intentionally smaller than the suggested example configuration because no GPU is available.



Selected epochs are tabulated below; all epochs and learning rates are available in results/history.csv.

Epoch	Training loss	Validation loss
1	7.9018	7.0632
3	6.7215	6.6720
5	6.4398	6.5012
7	6.2250	6.3722
9	6.0514	6.2745
11	5.9118	6.2062
12	5.8469	6.1827

4. Full-test quantitative evaluation

Greedy decoding begins with BOS, feeds predicted tokens back into the decoder, and stops at EOS or 128 generated tokens. PAD and BOS are blocked as output tokens. There is no reference input at generation time. Each article is encoded once; projected decoder keys and values are cached. Caching changes computation cost, not the greedy output (verified by tests).

Scores below are macro averages of per-example F1 multiplied by 100 on all 11,490 test pairs using rouge_score with use_stemmer=True. ROUGE-L uses one sequence; ROUGE-Lsum uses sentence-newline boundaries. Generated text and Lead-3 use an offline punctuation-based sentence heuristic; reference highlights preserve their existing newlines. This heuristic can split abbreviations incorrectly and is a limitation.

System	R-1	R-2	R-L	R-Lsum
Transformer	1.75	0.05	1.63	1.70
Lead-3 (measured)	39.91	17.38	24.90	36.20
Lead-3 (assignment)	40.06	17.48	25.02	36.36

Lead-3 copies the first three sentences of each full original article. It has access to the full article, whereas the model sees only its first 400 subwords. Its measured scores may differ from the assignment reference because sentence segmentation is heuristic. Both systems are scored with the identical local metric configuration.

Transformer minus measured Lead-3 ROUGE-1: -38.16 points. Learning fluent, factual generation from only 2,048 examples is much harder than copying informative news leads. The 4,096-piece vocabulary also splits rare names and numbers into many fragments, and source truncation can remove important details.

There are 4,190 distinct decoded predictions across the 11,490 inputs. Distinct outputs do not imply good summaries: repetition, omitted facts and unsupported statements must still be assessed. The original run collapsed to one output. The revised run reduces the learning-rate scale from 1.0 to 0.1 following validation-only diagnostics. The model architecture, training subset, tokenizer and seed are unchanged. The checkpoint is selected by validation loss, before final test scoring. The original test set has already been evaluated, so this is a reused benchmark, not a new blind held-out experiment. No test metric selects the revised hyperparameters or checkpoint.

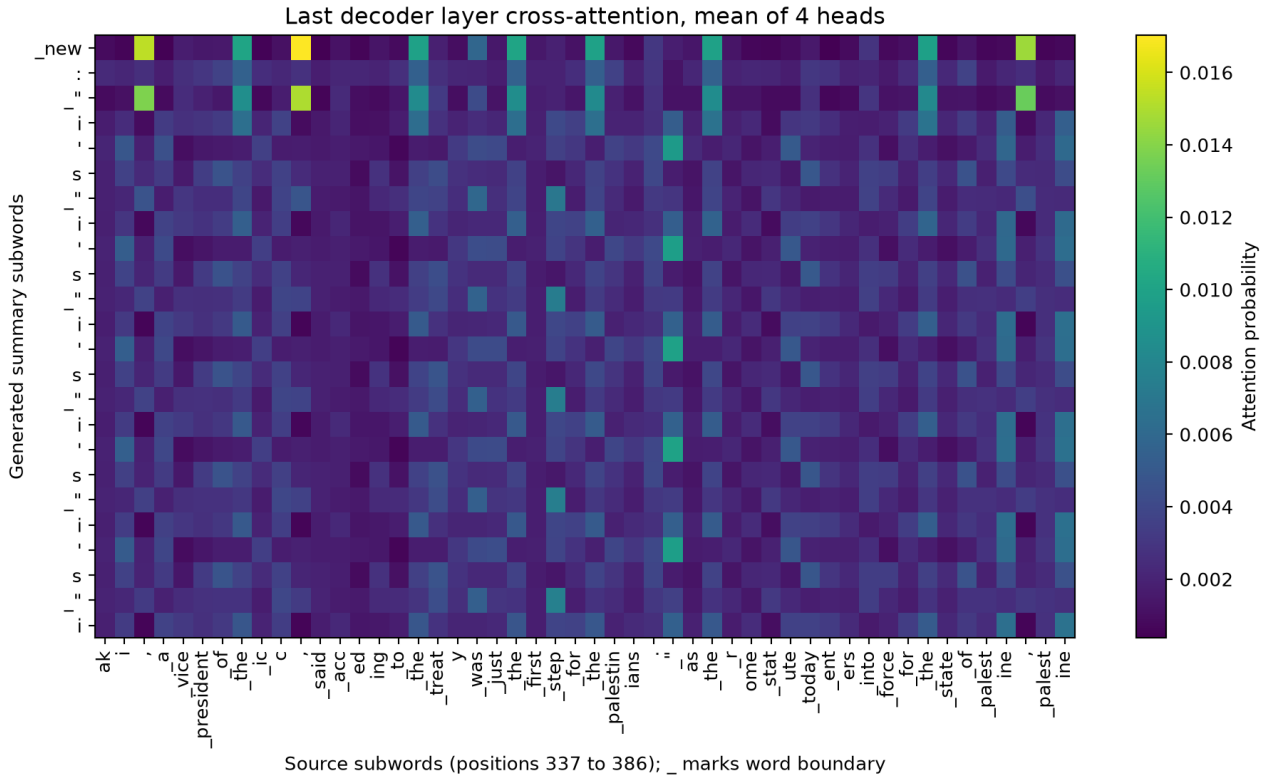
EOS-terminated outputs: 3,964/11,490. Outputs with any repeated whitespace trigram: 11,036/11,490. Generation plus scoring took 2.8 minutes. Repetition is measured on unedited decoded outputs; it is not corrected with sampling or repetition penalties.

Interpretation and limits

ROUGE measures reference overlap rather than factual consistency. Low cross-entropy alone cannot establish good summarization because teacher forcing supplies correct preceding words during validation. Greedy generation must recover from its own errors. This is one seed on a small, ordered subset, with a 1,024-example validation subset; no confidence intervals or claim of competitive summarization quality are made. More representative training data, a wider/deeper model, longer training, and additional validation would be the first improvements. Test scores must not guide those choices.

5. Attention recorded during generation

The heatmap uses cross-attention from the last decoder layer for test example 1. Four heads are averaged. Each row was recorded while generating its labelled predicted token; no gold target was fed to the model. The source window contains 50 consecutive subwords starting at index 337, chosen to maximize summed attention over all generated steps. For legibility the figure shows the first 24 generated subwords. Underscores mark SentencePiece word boundaries.



The selected window contains 13.6% of attention mass per generated row on average (over all generated rows). Probabilities are not renormalized after cropping. The complete attention matrix is supplied in results/attention.npz. A large weight indicates model focus, not proof that a source fact caused a correct prediction. The largest individual weight is 0.020997, compared with uniform 1/400 = 0.002500. A uniform 50-token window would hold 12.5% of the mass. Concentration alone does not establish a correct semantic alignment. Repeated rows alongside repeated output can indicate a decoding loop; diffuse attention can indicate weak alignment. Neither pattern alone proves an error mechanism.

6. Qualitative analysis protocol

The next five pages show the first five test examples in fixed repository order. This selection was fixed before evaluating outputs and avoids cherry-picking. Article excerpts are explicitly labelled; references and generated summaries are unedited and complete. A complete per-test prediction file is provided separately without articles. The diagnostics below each example describe repetition and overlap; they do not certify facts.

7.1. Test example 1

Dataset ID: f001ec5c4704938247d27a44948eebb37ae98d01

Article excerpt (first 1,100 characters)

(CNN)The Palestinian Authority officially became the 123rd member of the International Criminal Court on Wednesday, a step that gives the court jurisdiction over alleged crimes in Palestinian territories. The formal accession was marked with a ceremony at The Hague, in the Netherlands, where the court is based. The Palestinians signed the ICC's founding Rome Statute in January, when they also accepted its jurisdiction over alleged crimes committed "in the occupied Palestinian territory, including East Jerusalem, since June 13, 2014." Later that month, the ICC opened a preliminary examination into the situation in Palestinian territories, paving the way for possible war crimes investigations against Israelis. As members of the court, Palestinians may be subject to counter-charges as well. Israel and the United States, neither of which is an ICC member, opposed the Palestinians' efforts to join the body. But Palestinian Foreign Minister Riad al-Malki, speaking at Wednesday's ceremony, said it was a move toward greater justice. "As Palestine formally becomes a State Party to the Rome St

Reference highlights (original newlines)

Membership gives the ICC jurisdiction over alleged crimes committed in Palestinian territories since last June .
Israel and the United States opposed the move, which could open the door to war crimes investigations against Israelis .

Generated summary (unedited greedy output)

new: "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's" "i's" "i's" "i's "i's "i's" "i's "i's "i's" "i's" "i's"

Observed error indicators

The key reference information is ICC jurisdiction and Palestinian membership. Of the diagnostic source-name/topic strings icc, palestin, the output contains none. String presence alone does not establish a correct factual statement.

The unedited output contains 26 whitespace-delimited words; 16 repeated trigram occurrences beyond their first appearances. It shares 0 distinct whitespace tokens with the reference. These counts diagnose repetition and lexical overlap, not factual accuracy. Compare the source excerpt and reference above to check whether the output preserves the event, participants, and quantities. Generic or repeated text does not identify the key facts even when it contains common reference words.

7.2. Test example 2

Dataset ID: 230c522854991d053fe98a718b1defa077a8efef

Article excerpt (first 1,100 characters)

(CNN)Never mind cats having nine lives. A stray pooch in Washington State has used up at least three of her own after being hit by a car, apparently whacked on the head with a hammer in a misguided mercy killing and then buried in a field -- only to survive. That's according to Washington State University, where the dog -- a friendly white-and-black bully breed mix now named Theia -- has been receiving care at the Veterinary Teaching Hospital. Four days after her apparent death, the dog managed to stagger to a nearby farm, dirt-covered and emaciated, where she was found by a worker who took her to a vet for help. She was taken in by Moses Lake, Washington, resident Sara Mellado. "Considering everything that she's been through, she's incredibly gentle and loving," Mellado said, according to WSU News. "She's a true miracle dog and she deserves a good life." Theia is only one year old but the dog's brush with death did not leave her unscathed. She suffered a dislocated jaw, leg injuries and a caved-in sinus cavity -- and still requires surgery to help her breathe. The veterinary hospita

Reference highlights (original newlines)

Theia, a bully breed mix, was apparently hit by a car, whacked with a hammer and buried in a field .
"She's a true miracle dog and she deserves a good life," says Sara Mellado, who is looking for a home for Theia .

Generated summary (unedited greedy output)

new: "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i" "i's "i" "i's "i's "we" "i's "

Observed error indicators

The key reference information is Theia surviving severe injuries, her rescue and care. Of the diagnostic source-name/topic strings theia, dog, the output contains none. String presence alone does not establish a correct factual statement.

The unedited output contains 34 whitespace-delimited words; 23 repeated trigram occurrences beyond their first appearances. It shares 0 distinct whitespace tokens with the reference. These counts diagnose repetition and lexical overlap, not factual accuracy. Compare the source excerpt and reference above to check whether the output preserves the event, participants, and quantities. Generic or repeated text does not identify the key facts even when it contains common reference words.

7.3. Test example 3

Dataset ID: 4495ba8f3a340d97a9df1476f8a35502bcce1f69

Article excerpt (first 1,100 characters)

(CNN)If you've been following the news lately, there are certain things you doubtless know about Mohammad Javad Zarif. He is, of course, the Iranian foreign minister. He has been U.S. Secretary of State John Kerry's opposite number in securing a breakthrough in nuclear discussions that could lead to an end to sanctions against Iran -- if the details can be worked out in the coming weeks. And he received a hero's welcome as he arrived in Iran on a sunny Friday morning. "Long live Zarif," crowds chanted as his car rolled slowly down the packed street. You may well have read that he is "polished" and, unusually for one burdened with such weighty issues, "jovial." An Internet search for "Mohammad Javad Zarif" and "jovial" yields thousands of results. He certainly has gone a long way to bring Iran in from the cold and allow it to rejoin the international community. But there are some facts about Zarif that are less well-known. Here are six: . In September 2013, Zarif tweeted "Happy Rosh Hashanah," referring to the Jewish New Year. That prompted Christine Pelosi, the daughter of House Mino

Reference highlights (original newlines)

Mohammad Javad Zarif has spent more time with John Kerry than any other foreign minister .
He once participated in a takeover of the Iranian Consulate in San Francisco .
The Iranian foreign minister tweets in English .

Generated summary (unedited greedy output)

new: "i's "i's "i's "i's "i's "i" "i" "i's "i" "i" "i's "i's "i's "i" "i" "i" "i" obama"

Observed error indicators

The key reference information is Zarif, diplomacy with John Kerry, and English tweets. Of the diagnostic source-name/topic strings zarif, kerry, the output contains none. String presence alone does not establish a correct factual statement.

The unedited output contains 51 whitespace-delimited words; 38 repeated trigram occurrences beyond their first appearances. It shares 0 distinct whitespace tokens with the reference. These counts diagnose repetition and lexical overlap, not factual accuracy. Compare the source excerpt and reference above to check whether the output preserves the event, participants, and quantities. Generic or repeated text does not identify the key facts even when it contains common reference words.

7.4. Test example 4

Dataset ID: a38e72fed88684ec8d60dd5856282e999dc8c0ca

Article excerpt (first 1,100 characters)

(CNN)Five Americans who were monitored for three weeks at an Omaha, Nebraska, hospital after being exposed to Ebola in West Africa have been released, a Nebraska Medicine spokesman said in an email Wednesday. One of the five had a heart-related issue on Saturday and has been discharged but hasn't left the area, Taylor Wilson wrote. The others have already gone home. They were exposed to Ebola in Sierra Leone in March, but none developed the deadly virus. They are clinicians for Partners in Health, a Boston-based aid group. They all had contact with a colleague who was diagnosed with the disease and is being treated at the National Institutes of Health in Bethesda, Maryland. As of Monday, that health care worker is in fair condition. The Centers for Disease Control and Prevention in Atlanta has said the last of 17 patients who were being monitored are expected to be released by Thursday. More than 10,000 people have died in a West African epidemic of Ebola that dates to December 2013, according to the World Health Organization. Almost all the deaths have been in Guinea, Liberia and Si

Reference highlights (original newlines)

17 Americans were exposed to the Ebola virus while in Sierra Leone in March .
Another person was diagnosed with the disease and taken to hospital in Maryland .
National Institutes of Health says the patient is in fair condition after weeks of treatment .

Generated summary (unedited greedy output)

new: new: "i-old" new: "i-old, police say .
new: "i-old, police say .
new: "i-old, police say .
new: "i-old, police say .
new: "weed in the u.s.s.s

7.5. Test example 5

Dataset ID: c27cf1b136cc270023de959e7ab24638021bc43f

Article excerpt (first 1,100 characters)

(CNN)A Duke student has admitted to hanging a noose made of rope from a tree near a student union, university officials said Thursday. The prestigious private school didn't identify the student, citing federal privacy laws. In a news release, it said the student was no longer on campus and will face student conduct review. The student was identified during an investigation by campus police and the office of student affairs and admitted to placing the noose on the tree early Wednesday, the university said. Officials are still trying to determine if other people were involved. Criminal investigations into the incident are ongoing as well. Students and faculty members marched Wednesday afternoon chanting "We are not afraid. We stand together," after pictures of the noose were passed around on social media. At a forum held on the steps of Duke Chapel, close to where the noose was discovered at 2 a.m., hundreds of people gathered. "You came here for the reason that you want to say with me, 'This is no Duke we will accept. This is no Duke we want. This is not the Duke we're here to experi

Reference highlights (original newlines)

Student is no longer on Duke University campus and will face disciplinary review .
School officials identified student during investigation and the person admitted to hanging the noose, Duke says .
The noose, made of rope, was discovered on campus about 2 a.m.

Generated summary (unedited greedy output)

new: "we, police say .
new: "we, police say .
new: "we, police say .
new: "we, police say .
new: "we, police say .
new: "we, police say .
new: "we, police say .
new: "we, police say .
new: "we, police say .
new: "we, police say .
new: "we, police say .
new: "we, police say .
new: "we, police say .
new: "we, police say .
new: "we, police say .

Observed error indicators

The key reference information is Duke student disciplinary review following the noose incident. Of the diagnostic source-name/topic strings duke, noose, the output contains none. String presence alone does not establish a correct factual statement.

The unedited output contains 75 whitespace-delimited words; 68 repeated trigram occurrences beyond their first appearances. It shares 1 distinct whitespace tokens with the reference. These counts diagnose repetition and lexical overlap, not factual accuracy. Compare the source excerpt and reference above to check whether the output preserves the event, participants, and quantities. Generic or repeated text does not identify the key facts even when it contains common reference words.

8. Collapse investigation and correction

The original checkpoint was audited on validation indices 0, 1, 2, 100, 500 and 1000. All six source token sequences were different. Original encoder layer 1 retained differences (RMS 0.86-0.87), but layer 2 reduced them to about 0.0013. Fresh random weights produced encoder differences around 1.36-1.37. This is evidence of learned representation collapse rather than repeated tokenized input.

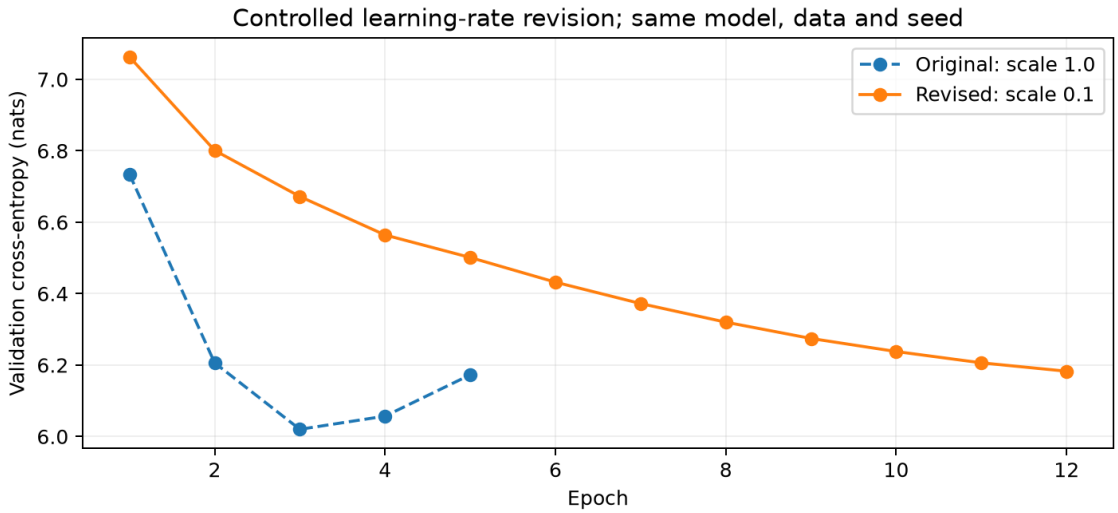
Cached and uncached greedy IDs matched exactly for all six real validation articles, including per-article versus batch runs and a shortened validation source to exercise padding. Hooks checked both decoder layers at every generation step: encoder K/V values and source masks were correct; blocked positions had zero attention. Cached versus full-prefix logit error was at most 5.7e-06. The complete original validation loss recomputed to exactly 6.0200552, matching the checkpoint. No decoder routing or cache defect was found.

The original schedule peaked at 0.0088388, which is aggressive for this small post-normalization model. A controlled retraining uses a 0.1 scale (peak 0.00088388), a maximum of 12 epochs and the same patience of 2. No architecture or decoding changes are introduced to force diverse outputs. The full audit is repeated on the revised validation checkpoint.

Validation diagnostic	Original	Revised
Distinct input sequences / 6	6	6
Distinct generated sequences / 6	1	6
Encoder RMS: article 1 vs 2	0.001306	1.445921
Cached = uncached	True	True
Recomputed validation loss	6.020055	6.182745

The original data/tokenizer reuse was valid, but the scripts failed to check it. The corrected scripts validate counts, unique/disjoint IDs, portable content hashes, BPE vocabulary, special token IDs and tokenizer fingerprints. Fresh streams from pinned revision 96df5e686bee6baa90b8bee7c28b81fa3fa6223d matched all three saved subsets. Training-only tokenizer reconstruction reproduced every piece and score.

9. Stability comparison



The comparison above uses validation only. The original run can have a lower teacher-forced loss while ignoring its source: the correct preceding target words still help next-token prediction. Restoring source sensitivity does not itself guarantee a lower loss, fluent generation or accurate facts. The revised checkpoint is chosen by the lowest validation loss within the revised, fixed training run. The unchanged small data sample remains a major limitation.

10. Verification and reproduction

Unit tests verify future-token isolation, source and target padding masking, attention normalization, ignored padding loss, summary shifting, sinusoidal positions, optimization on a fixed mini-batch, sentence boundaries, and equality of cached and full-prefix greedy predictions. See test_results.txt for the recorded run and README.md for commands.

Observed host: Python 3.14.8 on Windows 11; PyTorch runtime 2.14.1+cpu (distribution metadata 2.14.1), NumPy 2.5.4 in the local dependency overlay, datasets 5.1.0, SentencePiece 0.2.2, rouge-score 0.1.2, matplotlib 3.11.2, reportlab 5.1.0, nltk 3.10.4 and protobuf 7.36.2. The original training run used base NumPy 2.5.3; this revised run uses 2.5.4. PyTorch is unchanged. A separate virtual environment with these pinned direct dependencies and system-site-packages passed pip check, all 14 tests, dataset verification and the offline downloader reuse path. A fully isolated clean-machine installation is not claimed.

The archive includes source scripts, this report, measured metrics, training history, figures, five qualitative examples, test predictions and generation-time attention. It excludes dataset downloads, model checkpoints, the tokenizer training corpus and installed dependencies. Tokenizer and model artifacts are regenerated in a separate work directory.

References

Assignment: NLP Group Assignment 2, DSL 504, Question 3, pages 8-11.

Vaswani et al. (2017). Attention Is All You Need. https://arxiv.org/abs/1706.03762

CNN / Daily Mail 3.0.0, canonical Hugging Face repository: https://huggingface.co/datasets/abisee/cnn_dailymail

SentencePiece: https://github.com/google/sentencepiece

ROUGE implementation: https://github.com/google-research/google-research/tree/master/rouge

PyTorch softmax documentation: https://docs.pytorch.org/docs/stable/generated/torch.nn.Softmax.html

Plot-Twist Generator: RNN vs LSTM

Oct 30, 2026 · @HajmolaBurger

Summary

We trained a vanilla RNN and an LSTM language model on 20,000 TinyStories stories and used them to continue story openings. The LSTM is the better language model (test perplexity 13.40 vs 16.88) and loops less under greedy decoding, but neither model follows unusual prompts or produces a real plot twist.

Course: DSL 504 NLP, Group Assignment 2, Question 4. Seed 42 controls the data split and training; the Plot-Twist Challenge generator also sets the PyTorch seed to 42. The archived pooled metrics were produced by an earlier metrics script that did not explicitly seed generation; the revised script now does so on future runs.

Data and preprocessing

Dataset: `roneneldan/TinyStories` (Hugging Face Datasets). The loader streams the `train` split and takes the first 22,000 records from that stream. Those records are then shuffled with seed 42 and partitioned by story record into 20,000 training, 1,000 validation and 1,000 test records. The validation and test sets are therefore project-created held-out subsets of the Hugging Face `train` split; they are not the dataset publisher's official validation or test splits.

Split	Stories	Tokens	Avg length	<unk> rate	Truncated
Train	20,000	3,483,783	174.2	0.22%	32.7%
Val	1,000	173,213	173.2	0.34%	30.7%
Test	1,000	174,787	174.8	0.34%	32.2%

• Tokenisation: lowercase word-level regex, curly quotes normalised.

• Vocabulary: built from the train split only, minimum frequency 5, 6,695 words. Special tokens: <pad>=0, <unk>=1, <bos>=2, <eos>=3.

• Length: stories are cut at 200 tokens. Cut stories get no <eos>.

• Training pairs: each story is <bos> ... <eos>; the input is the sequence without its last token and the target is the sequence shifted by one.

• Why stories stay separate: records are split before sequences are created. Joining stories before splitting could create artificial transitions and allow one source story to contribute sequences to more than one split.

• Padding: batches are padded with <pad>, and the loss ignores pad positions (`ignore_index`), so padding never counts as an error.

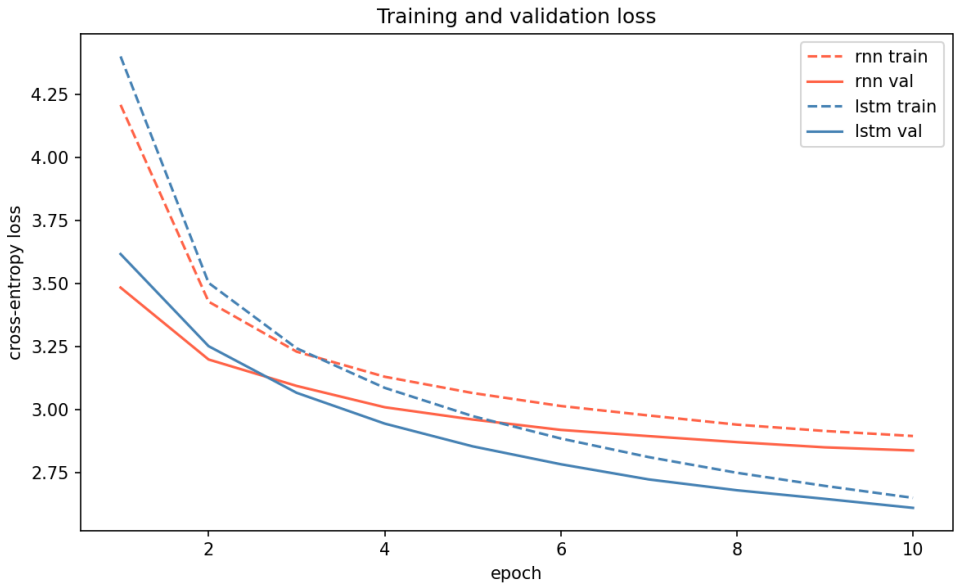
• Known issue: the cleaning left a few garbled characters (€, ™) in the vocabulary, and they show up in some generated text.

Models and training

Both models are Embedding → recurrent layer → Linear → vocabulary logits, built with `nn.RNN` and `nn.LSTM`. They share the data, vocabulary and settings below. The only difference is the recurrent layer.

Setting	Value (both models)
Embedding / hidden size	256 / 512
Layers	1
Dropout	0.2
Optimizer	Adam, learning rate 1e-3
Gradient clipping	1.0
Batch size	64
Max epochs / patience	10 / 3 (early stopping on validation loss)
Loss	Cross-entropy, padding ignored
Seed	42

Metric	RNN	LSTM
Parameters	5,542,695	6,725,415
Training time	344 s	473 s
Epochs run	10	10
Best validation loss	2.8372	2.6092
Validation perplexity	17.07	13.59
Test loss	2.8261	2.5955
Test perplexity	16.88	13.40



Training and validation loss for the RNN and LSTM

The LSTM's validation loss is below the RNN's from about epoch 3. Early stopping never triggered, and validation loss was still falling at epoch 10, so both models are undertrained. Training loss starts above validation loss because it is averaged over the epoch with dropout on. Training times depend on the hardware used.

Decoding

Strategy	How it picks the next word
Greedy	Always the most likely word
Temperature	Divide the scores by T, then sample. We used T = 0.5, 0.8 and 1.2
Top-k	Sample only from the k = 10 most likely words (T = 1.0)

Settings: at most 80 new tokens, stop at `<eos>`, `<pad>` and `<bos>` blocked, seed 42. The general prompts are in `src/generate.py` and the twist prompts in `src/challenge.py`. Every run is saved as a JSON file in `results/`.

Plot-Twist Challenge

We used seven twist openings: an exam paper with tomorrow's answers, a robot with a message from its future self, a detective solving a murder before it happens, a village forgetting the previous day, a letter from 100 years in the future, a new kid who knows everyone's secrets, and the moon moving closer to a village. Each was run with both models and three strategies (greedy, temperature 0.8, top-k 10), two samples each. That gives 84 records, but greedy is deterministic, so 70 are unique. Outputs were not edited.

We judged each output on four things: connection to the prompt, a meaningful twist, consistency, and repetition.

Model, strategy	Prompt	Output (unedited)	Verdict
RNN, greedy	A detective solved a murder that had not happened yet.	he was very excited and wanted to play with it . he asked his mom , " what is that ? " the bird said , " i am a little girl . i am a little girl . i am a little girl . i am a little girl ...	Loses the prompt, then loops

Model, strategy	Prompt	Output (unedited)	Verdict
RNN, greedy	The new kid at school knew everyone's secrets but had never met them.	he was so happy and he had a great time .	Ends early, no twist
LSTM, temperature 0.8	The new kid at school knew everyone's secrets but had never met them.	it was the best day ever !	Ends early, no twist
RNN, top-k	An old robot received a message from its future self.	they were going to play and play together . the end of its friends and they were both very tired . they had a fun time in the park and the sun was shining , and he was very happy . he had so much fun playing in the sand , but it was too late . the end .	Fluent but ignores the robot and the message
LSTM, greedy	Every night, the moon moved closer to the village.	the sun was shining and the sky was shining . the sun was shining brightly and the sky was full of bright colors . the sun was shining and the sky was so big and strong ...	Same idea repeated, moon forgotten
LSTM, temperature 0.8	A letter arrived from one hundred years in the future.	it was a fun day and the west said € hello ! it was really nice . let € ™ s go back and see what is there ! ...	Broken characters, no twist

(Long outputs are cut with "..." in this table only. The full text is in `results/plot_twist_challenge.json`.)

What we saw

• Relevance: almost every output drifts to a generic children's scene (a girl, a bird, a tree, a park). None that we found turned the premise into a real twist.

• Consistency: characters change mid-story (for example a robot becomes a little boy).

• Repetition: greedy decoding looped in 4 of 7 RNN outputs and 2 of 7 LSTM outputs.

• Failures kept in: loops, one-sentence endings, <unk> tokens and broken characters.

Quantitative results

Higher diversity is not the same as a better story. Only the generated text is measured here (prompt removed). Distinct-1/2 is unique unigrams/bigrams divided by total. Repetition is 1 − unique 3-grams / total 3-grams, averaged per output. A loop is an output with repetition above 0.3.

By strategy (Plot-Twist Challenge, unique outputs)

Model	Strategy	Outputs	Distinct-1	Distinct-2	Repetition	Loops
RNN	Greedy	7	0.166	0.308	0.363	4
RNN	Temperature 0.8	14	0.339	0.754	0.002	0
RNN	Top-k (10)	14	0.286	0.723	0.003	0
LSTM	Greedy	7	0.197	0.420	0.211	2
LSTM	Temperature 0.8	14	0.345	0.771	0.012	0
LSTM	Top-k (10)	14	0.294	0.710	0.021	0

Temperature sweep (7 general prompts, one output each)

Model	Setting	Distinct-1	Distinct-2	Repetition
RNN	Greedy	0.106	0.184	0.528
RNN	Temperature 0.5	0.350	0.706	0.072
RNN	Temperature 1.2	0.560	0.964	0.000
LSTM	Temperature 0.5	0.315	0.669	0.067
LSTM	Temperature 0.8	0.453	0.824	0.031
LSTM	Temperature 1.2	0.575	0.962	0.004

The two tables use different prompt sets, so compare within a table only.

Pooled metrics (35 outputs per model: 7 general prompts × 5 settings; the original length measure includes the prompt)

Metric	RNN	LSTM
Test perplexity	16.88	13.40
Distinct-1	0.189	0.205
Distinct-2	0.480	0.544
Mean 3-gram repetition	0.103	0.084
Average full-output length (words, prompt + continuation)	70.8	78.9

The length values above are not continuation-only lengths. The generation function returns the prompt followed by the generated continuation, and the original metrics script counted words in that combined string. Therefore, 70.8 and 78.9 words include the opening prompt.

The original `under 20 words` count also used the combined prompt-plus-continuation text. It cannot reliably identify short or prematurely ended continuations because the prompt contributes 8–12 words before generation begins. It should not be interpreted as evidence that no continuation ended early.

Supplemental continuation-only lengths from the saved Plot-Twist Challenge outputs

To provide a prompt-excluded measurement without mixing sample sets, the following values were recomputed from all 84 saved challenge records (42 per model; 7 challenge prompts × 3 strategies × 2 samples). The exact prompt prefix was removed from each output before counting words with the same word-like-token pattern used by the metrics script. The two greedy samples for a given prompt are identical because greedy decoding is deterministic.

Model	Challenge records	Mean continuation words	Median	Range	Continuations under 20 words
RNN	42	62.9	67.0	10–73	2
LSTM	42	68.4	70.5	6–74	1

These 84 challenge records are a different sample set from the 35 general-prompt outputs used in the pooled metrics table. The supplementary length figures must therefore be read separately, not combined with or substituted into that table. A continuation shorter than 20 words is only a length flag; it does not by itself prove that the model terminated prematurely or that the output is incomplete.

Length-threshold limitation. The generation cap is 80 new tokens. The original `180+ words` threshold is above the maximum practically reachable length for these short prompts, even when the prompt is included. Its zero count is expected by construction and is not evidence that the model avoids excessively long outputs. The generation code also does not save a stop-reason field, so the existing records do not distinguish stopping at `<eos>` from reaching the token cap.

The perplexity, diversity and repetition values above are retained from the archived `artifacts/metrics.json`; they were not regenerated during this report correction. The updated `src/metrics.py` now reports continuation-only length on the next run and seeds its sampling, but rerunning it would produce a new stochastic generation set. Each setting has few samples and one seed, so small differences are not reliable.

Human evaluation

Method. 30 stories were sampled from the Plot-Twist Challenge outputs and shuffled (seed 42) as S01–S30. The CSV shows only the story text and sample ID, not the model or decoding strategy. The prepared rating sheet asks evaluators to score each sample from 1–5 for coherence, grammar, relevance to the prompt, and creativity/surprise. The key that links sample IDs to model and strategy is kept in a separate file. The 30 stories are not balanced: LSTM has 17 (temperature 6, top-k 6, greedy 5) and RNN has 13 (top-k 6, temperature 4, greedy 3).

Who rated. Not yet collected. All four rating fields for each of the 30 samples in `results/human_eval_blinded.csv` are blank. No evaluator count or human-rating result can be reported at this stage.

Results. Pending actual evaluator responses. The table below is intentionally left blank; no scores have been fabricated or inferred.

Criterion	RNN average	LSTM average
Coherence	–	–
Grammar	–	–
Relevance	–	–
Creativity / surprise	–	–

Disagreement between raters and with perplexity. Pending actual ratings. It is not yet possible to calculate evaluator disagreement or compare human scores with the lower LSTM perplexity.

Dataset split provenance and run environment

What the code establishes. `src/data.py` calls `load_dataset("roneneldan/TinyStories", split="train", streaming=True)`, takes the first 22,000 records from that stream, and shuffles those records using `random.Random(42)`. It then slices the shuffled list into 20,000 training, 1,000 validation and 1,000 test story records. Tokenization and conversion into next-token sequences happen after this partitioning. The three project splits are thus disjoint slices of the selected record list by construction; the official Hugging Face validation and test splits are not used.

Leakage status. The inspected code does not demonstrate actual train/validation/test leakage: it partitions story records before creating token sequences and does not intentionally reuse a list entry across splits. However, the archive does not retain upstream story IDs, a dataset revision/commit, or an exact/near-duplicate audit. Duplicate or near-duplicate text appearing as separate source records therefore cannot be ruled out. The defensible conclusion is that leakage has not been established from the available evidence; it would be inaccurate either to claim that leakage occurred or to claim that every possible form of duplicate leakage was ruled out.

Dataset revision and provenance limits. The dataset identifier is recorded as `roneneldan/TinyStories`, but no Hugging Face revision/commit hash or streamed-record IDs are saved. The exact first 22,000 records may consequently depend on the upstream dataset revision and streaming order available when preprocessing ran. `artifacts/data_stats.json` preserves aggregate split sizes and token statistics, not source record identifiers.

Original runtime environment. The code selects CUDA when `torch.cuda.is_available()` is true and otherwise uses CPU. The saved training histories contain the seed, parameter count, loss/perplexity history and elapsed training seconds, but do not identify the actual device, GPU/CPU model, Python version, PyTorch version, CUDA version, or `datasets` version. `requirements.txt` lists package names without version pins. `COLAB_STEPS.md` describes a suggested T4-GPU workflow; it is not a record that the original training run used a T4. The original runtime environment is therefore only partially documented and cannot be reconstructed from this archive alone. Add exact hardware/software versions only from the original notebook or run logs; do not infer them from the instructions.

Final analysis

1. How do RNN and LSTM differ? The LSTM predicts text better: test perplexity 13.40 vs 16.88. Its gates help it keep information longer. It costs about 21% more parameters and 38% more training time. Both were still improving at epoch 10.

2. Which is more coherent and less repetitive? The LSTM is less repetitive, mostly under greedy decoding (repetition 0.211 vs 0.363; 2 vs 4 looping outputs out of 7). Both ignore the unusual prompts and write generic stories. Human coherence ratings have not yet been collected; the Human evaluation section remains pending.

3. How does temperature affect creativity and grammar? Higher temperature gives more varied words but worse grammar. At 0.5 the text is readable but samey. At 0.8 it is a good balance. At 1.2 it falls apart, for example: "it waited what was jane standing in the blue - <unk> sparkling coming water".

4. Do high diversity scores guarantee better stories? No. Temperature 1.2 has the highest diversity (Distinct-2 about 0.96) and the worst text. Diversity counts unique words, not whether they make sense.

5. Does lower perplexity mean better storytelling? Only a little. The LSTM loops less and reads more smoothly, but neither model follows the prompt or makes a real twist. The human-rating comparison remains undetermined because the rating fields are still blank.

6. What failures are most common, and how could they be reduced?

• Ignoring the prompt: more data and training, or conditioning on the prompt.

• Repetition loops (greedy): a repetition penalty, or sampling instead of greedy.

• Characters changing mid-story: a bigger model and longer training.

• Gibberish at high temperature: use temperature 0.8 or top-k.

• Broken characters and <unk>: fix the text cleaning and retrain.

These fixes were not tested in our experiments.

7. Limitations

• Small data: 20,000 stories and a word-level vocabulary of 6,695 words.

• Undertrained: 10 epochs and one seed (42).

• Stories cut at 200 tokens and outputs cut at 80 tokens.

• The models predict one word at a time and do not plan a plot, so real twists are unlikely.

• Small evaluation: 7 prompts and few samples per setting. Thirty blinded rating rows are prepared, but their rating cells are blank, so human evaluation is not yet complete.

Conclusion

The LSTM is the stronger language model on the recorded test perplexity (13.40 vs 16.88) and has a lower measured repetition rate. Temperature 0.8 or top-k gave the more readable examples in the inspected samples, while very high temperature produced gibberish and greedy decoding often looped. Neither model reliably turned the twist prompts into meaningful plot twists, so lower perplexity and higher diversity should not be read as proof of better storytelling. Both models are undertrained, the results come from one seed and small samples, and human ratings remain uncollected.

To reproduce: see the README. Code, checkpoints, vocabulary and all generated outputs are in the submitted archive.

Appendix: AI-simulated proxy evaluation

Plot-Twist Generator — Evaluation Analysis

Evaluation dataset and provenance

Thirty blinded samples (S01–S30) were rated in three AI-simulated passes, producing 90 records. The three passes are rubric perspectives from one AI-assisted workflow, not independent human evaluators. The ratings are used as a proxy dataset for the requested descriptive evaluation; they must not be represented as genuine human ratings.

Criterion means across all 90 ratings

Criterion	Mean (1–5)
Coherence	1.83
Grammar	2.98
Prompt relevance	1.00
Creativity/surprise	1.40
Overall mean	1.80

Means by AI-simulated rating pass

Pass	Coherence	Grammar	Relevance	Creativity/surprise
AI-Simulated-R1	1.83	2.67	1.00	1.07
AI-Simulated-R2	1.83	3.60	1.00	1.07
AI-Simulated-R3	1.83	2.67	1.00	2.07

Disagreement analysis

Disagreement is reported as the range across the three passes for each sample and criterion. This is descriptive variation between AI rubric passes, not human inter-rater reliability.

Criterion	Samples with any disagreement	Maximum range	Exact three-way agreement
Coherence	0/30	0	100.0%
Grammar	28/30	1	6.7%
Prompt relevance	0/30	0	100.0%
Creativity/surprise	30/30	1	0.0%

Grammar differed in 28/30 samples, while creativity/surprise differed in all 30; each observed range was at most one point. Coherence and relevance were identical across the three passes for every sample. The prompt-fit/novelty pass assigned a higher creativity/surprise mean (2.07) than the other two passes (1.07 each).

Qualitative synthesis

• Prompt drift was the dominant weakness: continuations often abandoned the opening premise.

• Local continuity was weak, with unrelated characters, objects, and settings introduced without explanation.

• Grammar/readability was stronger than narrative quality, although awkward dialogue, punctuation artifacts, and unclear references appeared.

• Repetition often replaced plot progression and meaningful surprise.

• S14 and S22 were identified as locally more understandable, but neither meaningfully resolved its opening premise.

Perplexity comparison

Archived test perplexity is 16.88 for the Vanilla RNN and 13.40 for the LSTM (lower is better). The proxy rating averages indicate weak narrative quality overall, particularly prompt relevance (1.00/5), despite the LSTM's lower perplexity. This is a qualitative juxtaposition only: sample-level perplexities and a verified mapping from S01–S30 to model are unavailable. No correlation coefficient or model-specific rating comparison is justified. Lower perplexity measures next-token prediction, not story relevance or plot quality.

Blinding and provenance

No RNN/LSTM or decoding-strategy labels were inferred for S01–S30. These are AI-simulated proxy ratings, not genuine human ratings. They should not be described as three independent human evaluators or as evidence of human inter-rater reliability.

Experimental summary and limitations

Question	Experiment and result	Limitations
Q1: translation	50,000 training pairs; LSTM baseline versus additive attention; test BLEU 0.60 versus 0.92	One seed per model; word-level vocabulary; filtered test subset; low translation quality
Q2: review ratings	200,000 training reviews; two seeds; test Macro F1 0.6351 for BiGRU versus 0.6706 for RoBERTa	Two seeds; sequence truncation; mixed training topology; pretrained versus non-Transformer comparison
Q3: summarization	2,048 training examples; 11,490 test articles; revised model ROUGE-1 1.7506 versus Lead-3 39.9077	Small ordered subset; pervasive repetition; reused test benchmark; no competitive quality claim
Q4: story generation	20,000 training stories; RNN/LSTM test perplexity 16.88/13.40; seven challenge prompts	Prompt drift and repetition; pooled metrics include prompts; AI-simulated proxy results are labelled separately; human sheets remain blank as stated by the authors

Q4 length measurements use separate sample sets. The original pooled metrics summarize 35 general-prompt outputs per model and include prompt words. The latest report separately measures all 84 challenge records after removing the exact prompt: 42 records per model, with mean continuation lengths 62.9 words (RNN) and 68.4 (LSTM); two and one continuations respectively are shorter than 20 words. These are descriptive length flags, not proof of incomplete stories. The authors note that an 180-word threshold is unreachable under their 80-new-token cap.

Across the experiments, attention and pretraining improve some measured tasks but do not guarantee fluent generation, factual accuracy or narrative coherence. Next-token loss and overlap/diversity metrics address different aspects of performance. Qualitative examples and explicit error analysis are needed to interpret the numerical results.

The supplied source, configurations, executed notebooks, predictions, figures and model-specific report sections document the experiment settings. Reproduction instructions are provided in the main README and each question directory. Dataset/model revisions and package versions are recorded where available; numerical equivalence across environments is not guaranteed.